

Lectures 3 and 4: The Central Limit Theorem

Swapping, the Lindeberg condition, and Stein's method

OUTLINE maximum entropy $\rightarrow$ why a Gaussian at all $\rightarrow$ what a limit in distribution is $\rightarrow$ swap one summand at a time $\rightarrow$ the CLT, and Lindeberg's condition

$\mathbb{E}[Wf(W)] = \mathbb{E}[f'(W)] \rightarrow$ vector and matrix forms $\rightarrow$ Stein's equation $\rightarrow$ the CLT with a rate $\rightarrow$ the same argument for Poisson

HOW TO READ Sections marked * are not lectured. They are complete and they are examinable. Exercises are typed A, B and C; each week's quiz has one of each. What is collected is listed in a box at the end of each section; everything else is a pool — available and examinable, not collected.

1 Why Gaussians appear, and the CLT by swapping

Q. Why should the Gaussian, of all laws, be the one that appears?

1.1 Notation

Throughout, $X_1, X_2, \dots$ are independent and identically distributed with mean 0 and variance 1, and

$$S_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i.$$

The normalization is chosen so that $\text{Var}(S_n) = 1$ for every n . We write $g \sim \mathcal{N}(0, 1)$, and γ for its law.

From §1.5* of Lectures 1–2. For a random variable X on $\mathbb{R}$ with density p , the *differential entropy* is $h(X) = -\int p \log p$, and the *relative entropy* of X from γ is $D(X\|\gamma) = \int p \log(p/\phi)$, where ϕ is the standard normal density. Both may be $+\infty$, and h is undefined when X has no density.

1.2 What “converging to a Gaussian” means

S_n does not converge to a random variable; what converges is its law, and we test that against functions.

Definition 1.1 (Convergence in distribution). For random vectors in $\mathbb{R}^k$, $X_n \Rightarrow X$ means $\mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X)$ for every bounded continuous $f : \mathbb{R}^k \rightarrow \mathbb{R}$.

Distribution functions need not converge at a jump: if $X_n \equiv 1/n$ and $X \equiv 0$ then $X_n \Rightarrow X$, yet $F_n(0) = 0$ for every n while $F(0) = 1$.

A smooth f can be Taylor expanded and $\mathbf{1}_{(-\infty, z]}$ cannot, so the proofs below test against smooth functions only. That is already enough.

Lemma 1.2. Suppose $\mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X)$ for every $f \in C^3(\mathbb{R})$ with f, f', f'', f''' bounded. Then $F_{X_n}(z) \rightarrow F_X(z)$ at every z where F_X is continuous.

Proof. Fix such a z and $\delta > 0$. Let $\psi : \mathbb{R} \rightarrow [0, 1]$ be smooth with $\psi = 1$ on $(-\infty, 0]$ and $\psi = 0$ on $[1, \infty)$, and put

$$u^+(x) = \psi\left(\frac{x-z}{\delta}\right), \quad u^-(x) = \psi\left(\frac{x-z+\delta}{\delta}\right).$$

Both lie in the test class, and

$$\mathbf{1}_{(-\infty, z-\delta]} \leq u^- \leq \mathbf{1}_{(-\infty, z]} \leq u^+ \leq \mathbf{1}_{(-\infty, z+\delta]}.$$

Taking expectations at X_n and passing to the limit,

$$F_X(z - \delta) \leq \mathbb{E}u^-(X) = \lim_n \mathbb{E}u^-(X_n) \leq \liminf_n F_{X_n}(z),$$

$$\limsup_n F_{X_n}(z) \leq \lim_n \mathbb{E}u^+(X_n) = \mathbb{E}u^+(X) \leq F_X(z + \delta).$$

Let $\delta \downarrow 0$ and use continuity of F_X at z . ■

A rate against smooth f is still not a rate against probabilities; §2.3 says what converting one into the other costs.

Exercise 1.3 (TYPE C). Let X_n, X be random vectors in $\mathbb{R}^k$ with $\mathbb{E}f(X_n) \rightarrow \mathbb{E}f(X)$ for every $f \in C^3(\mathbb{R}^k)$ whose derivatives up to order three are bounded. Show $X_n \Rightarrow X$. Such an f is neither smooth nor compactly supported; repair each defect in turn, and note that the hypothesis already limits how much mass X_n can put far out.

1.3 The moral answer

Section 1.5* of Lectures 1–2 supplies two facts. First, at fixed variance the Gaussian maximizes entropy: $h(X) \leq h(g)$ whenever $\text{Var}(X) = 1$. Second, for symmetric i.i.d. X, Y ,

$$h\left(\frac{X+Y}{\sqrt{2}}\right) \geq h(X),$$

because $(x, y) \mapsto \left(\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}}\right)$ is orthogonal, orthogonal maps preserve differential entropy, and entropy is subadditive.

Since $S_{2^k} = \frac{1}{\sqrt{2}}(S'_{2^{k-1}} + S''_{2^{k-1}})$ for two independent copies, $h(S_{2^k})$ increases in k and never exceeds $h(g)$. So it converges, and the maximizer is the only candidate for the limit.

1.4 The entropic CLT

Theorem 1.4 (Barron, 1986). *If $D(S_m \parallel \gamma) < \infty$ for some m , then $D(S_n \parallel \gamma) \rightarrow 0$; equivalently $h(S_n) \rightarrow h(g)$.*

Theorem 1.5 (Artstein–Ball–Barthe–Naor, 2004). *$h(S_n)$ is increasing in n , not merely along powers of two.*

We will not prove either.

Example 1.6 (The entropic CLT says nothing about a coin flip). Let $X_i = \pm 1$ with probability $\frac{1}{2}$ each. Then S_n is supported on a finite set for every n , so it has no density, $h(S_n)$ is undefined, and $D(S_n \parallel \gamma) = \infty$ for every m and n . The hypothesis of Barron’s theorem never holds.

The classical central limit theorem holds for ± 1 regardless, so we need a proof that covers Example 1.6.

1.5 Lindeberg's swapping argument

Theorem 1.7 (Central limit theorem). *Suppose in addition $\mathbb{E}|X_1|^3 < \infty$. Then for every $f \in C^3(\mathbb{R})$ with bounded third derivative,*

$$|\mathbb{E}f(S_n) - \mathbb{E}f(g)| \leq \frac{\|f'''\|_\infty}{6\sqrt{n}} \left(\mathbb{E}|X_1|^3 + \mathbb{E}|g|^3 \right).$$

Proof. Let $g_1, \dots, g_n$ be i.i.d. standard Gaussians, independent of the X_i , and interpolate:

$$Z_k = \frac{1}{\sqrt{n}}(g_1 + \dots + g_k + X_{k+1} + \dots + X_n), \quad k = 0, \dots, n,$$

so $Z_0 = S_n$ and $Z_n \sim \mathcal{N}(0, 1)$. Telescoping,

$$\mathbb{E}f(S_n) - \mathbb{E}f(g) = \sum_{k=1}^n \left(\mathbb{E}f(Z_{k-1}) - \mathbb{E}f(Z_k) \right).$$

Fix k and write $W_k = \frac{1}{\sqrt{n}}(g_1 + \dots + g_{k-1} + X_{k+1} + \dots + X_n)$, which is independent of both X_k and g_k . Then $Z_{k-1} = W_k + X_k/\sqrt{n}$ and $Z_k = W_k + g_k/\sqrt{n}$, and Taylor's theorem gives

$$f(W_k + u) = f(W_k) + uf'(W_k) + \frac{u^2}{2}f''(W_k) + R(u), \quad |R(u)| \leq \frac{\|f'''\|_\infty}{6}|u|^3.$$

Take expectations of both expansions and subtract. The constant terms cancel. The linear terms cancel because W_k is independent of X_k and g_k and both have mean 0. The quadratic terms cancel because both have variance 1. Only the remainders survive:

$$|\mathbb{E}f(Z_{k-1}) - \mathbb{E}f(Z_k)| \leq \frac{\|f'''\|_\infty}{6n^{3/2}} \left(\mathbb{E}|X_k|^3 + \mathbb{E}|g_k|^3 \right).$$

Summing the n terms gives the claim. ■

The proof used Taylor's theorem and nothing else; in particular it never objected to $X_i = \pm 1$.

With Lemma 1.2, $\mathbb{P}(S_n \leq z) \rightarrow \Phi(z)$ for every z , where Φ is the standard normal distribution function.

The classical argument — expand $\varphi_{X_1}(t/\sqrt{n})$ and take n th powers — is carried out in §1.9*.

1.6 What the proof actually needed

The argument never used that the X_i share a law, nor that they are one-dimensional: the same swap with multivariate Taylor gives the multidimensional central limit theorem. What it did use is that no single summand carries much of the variance.

Definition 1.8 (Lindeberg's condition). Let $\{X_{n,i}\}_{i \leq k_n}$ be a triangular array, independent within rows, with

$$\mathbb{E}X_{n,i} = 0, \quad s_n^2 = \sum_{i \leq k_n} \text{Var}(X_{n,i}), \quad k_n \rightarrow \infty, \quad s_n \rightarrow \infty.$$

The array satisfies *Lindeberg's condition* if for every $\varepsilon > 0$

$$L_n(\varepsilon) := \frac{1}{s_n^2} \sum_i \mathbb{E} \left[X_{n,i}^2 \mathbf{1}\{|X_{n,i}| > \varepsilon s_n\} \right] \rightarrow 0.$$

Normalize once and for all:

$$Y_{n,i} = \frac{X_{n,i}}{s_n}, \quad \sigma_{n,i}^2 = \text{Var}(Y_{n,i}), \quad T_n = \sum_{i \leq k_n} Y_{n,i},$$

so that $\sum_i \sigma_{n,i}^2 = 1$ and $L_n(\varepsilon) = \sum_i \mathbb{E}[Y_{n,i}^2 \mathbf{1}\{|Y_{n,i}| > \varepsilon\}]$.

Theorem 1.9 (Lindeberg). *Under Lindeberg's condition, $T_n \Rightarrow \mathcal{N}(0, 1)$.*

Proof. Let $f \in C^3(\mathbb{R})$ with f'' and f''' bounded. Taylor's theorem to second and to third order gives, uniformly in w ,

$$f(w+u) = f(w) + uf'(w) + \frac{u^2}{2}f''(w) + R(u), \quad |R(u)| \leq \min\left(\frac{\|f'''\|_\infty}{6}|u|^3, \|f''\|_\infty u^2\right), \quad (1)$$

the second bound from $R(u) = \frac{u^2}{2}(f''(w+\theta u) - f''(w))$ for some $\theta \in (0, 1)$. It costs a power of u and buys freedom from the third derivative, which is what a summand with a heavy tail needs.

Splitting the second moment of $Y_{n,i}$ at ε gives, for every i and every $\varepsilon > 0$,

$$\sigma_{n,i}^2 = \mathbb{E}[Y_{n,i}^2 \mathbf{1}\{|Y_{n,i}| \leq \varepsilon\}] + \mathbb{E}[Y_{n,i}^2 \mathbf{1}\{|Y_{n,i}| > \varepsilon\}] \leq \varepsilon^2 + L_n(\varepsilon), \quad (2)$$

so $\limsup_n \max_i \sigma_{n,i}^2 \leq \varepsilon^2$ for every ε , that is $\max_i \sigma_{n,i} \rightarrow 0$.

Let $g_{n,i} \sim \mathcal{N}(0, \sigma_{n,i}^2)$ be independent of each other and of the array, so that $\sum_i g_{n,i} \sim \mathcal{N}(0, 1)$, and swap one summand at a time as in Theorem 1.7: with

$$W_{n,i} = g_{n,1} + \cdots + g_{n,i-1} + Y_{n,i+1} + \cdots + Y_{n,k_n},$$

which is independent of $Y_{n,i}$ and of $g_{n,i}$, the telescoping sum gives

$$|\mathbb{E}f(T_n) - \mathbb{E}f(g)| \leq \sum_i \left| \mathbb{E}f(W_{n,i} + Y_{n,i}) - \mathbb{E}f(W_{n,i} + g_{n,i}) \right|.$$

Expand both arguments by (1). The constant, linear and quadratic terms cancel term by term, because $Y_{n,i}$ and $g_{n,i}$ have the same first two moments and both are independent of $W_{n,i}$; only remainders survive. On the Y side use the cubic bound where $|Y_{n,i}| \leq \varepsilon$ and the quadratic one where $|Y_{n,i}| > \varepsilon$:

$$\sum_i \mathbb{E}|R(Y_{n,i})| \leq \frac{\|f'''\|_\infty}{6} \varepsilon \sum_i \mathbb{E}[Y_{n,i}^2] + \|f''\|_\infty \sum_i \mathbb{E}[Y_{n,i}^2 \mathbf{1}\{|Y_{n,i}| > \varepsilon\}] = \frac{\|f'''\|_\infty}{6} \varepsilon + \|f''\|_\infty L_n(\varepsilon).$$

On the Gaussian side there is nothing to truncate: $\mathbb{E}|g_{n,i}|^3 = 2\sqrt{2/\pi} \sigma_{n,i}^3$ and $\sum_i \sigma_{n,i}^3 \leq \max_i \sigma_{n,i}$, so

$$\sum_i \mathbb{E}|R(g_{n,i})| \leq \frac{\|f'''\|_\infty}{3} \sqrt{\frac{2}{\pi}} \max_i \sigma_{n,i}.$$

Adding the two and letting $n \rightarrow \infty$: $L_n(\varepsilon) \rightarrow 0$ by hypothesis and $\max_i \sigma_{n,i} \rightarrow 0$ by (2), so

$$\limsup_{n \rightarrow \infty} |\mathbb{E}f(T_n) - \mathbb{E}f(g)| \leq \frac{\|f'''\|_\infty}{6} \varepsilon$$

for every $\varepsilon > 0$, and the left side is 0. Lemma 1.2 converts this into convergence in distribution. $\blacksquare$

Remark 1.10 (Feller's condition). Say the array satisfies *Feller's condition* if

$$\max_{i \leq k_n} \frac{\text{Var}(X_{n,i})}{s_n^2} = \max_{i \leq k_n} \sigma_{n,i}^2 \longrightarrow 0.$$

Three precise statements relate it to Definition 1.8.

- (a) Lindeberg's condition implies Feller's. This is exactly (2), and it is the only place the proof above used anything beyond $L_n(\varepsilon) \rightarrow 0$.
- (b) The implication is strict: Exercise 1.14 exhibits an array satisfying Feller's condition and not Lindeberg's.
- (c) *Converse, stated without proof.* If Feller's condition holds and $T_n \Rightarrow \mathcal{N}(0, 1)$, then Lindeberg's condition holds.

Together with Theorem 1.9, (c) says that among arrays with no dominant summand Lindeberg's condition is not merely sufficient for the central limit theorem but equivalent to it. The pair of statements is the *Lindeberg–Feller theorem*.

Remark 1.11 (Lyapunov's condition). In practice one checks a cruder condition. If for some $\delta > 0$

$$\frac{1}{s_n^{2+\delta}} \sum_i \mathbb{E}|X_{n,i}|^{2+\delta} \longrightarrow 0,$$

then Lindeberg's condition holds: on $\{|X_{n,i}| > \varepsilon s_n\}$ one may write $X_{n,i}^2 \leq |X_{n,i}|^{2+\delta}/(\varepsilon s_n)^\delta$. For a weighted sum $X_{n,i} = a_{n,i}\xi_i$ of i.i.d. noise with $\mathbb{E}|\xi_1|^3 < \infty$ this reads

$$L_n(\varepsilon) \leq \frac{\mathbb{E}|\xi_1|^3}{\varepsilon} \left(\frac{\|a_n\|_3}{\|a_n\|_2} \right)^3,$$

so such a sum is asymptotically Gaussian as soon as the weight vector is delocalized. A vector with a bounded number of dominant entries is exactly what this excludes.

Lindeberg's condition separates two regimes: many small contributions give a Gaussian, a few rare ones give a Poisson. Section 1.9* computes both sides of the boundary; §2.4 proves the Poisson statement, with a rate.

Remark 1.12 (The limit is universal, the rate is not). Kolmogorov–Smirnov distance from S_{30} to $\mathcal{N}(0, 1)$, by simulation: uniform 0.0014, exponential 0.025, ± 1 0.073. The limit does not depend on the law of X_1 ; the speed does, and lattice summands are slowest.

Exercise 1.13 (TYPE A). Let $X_1, X_2, \dots$ be i.i.d. with mean 0 and variance $\sigma^2 \in (0, \infty)$, and set $X_{n,i} = X_i$ for $i \leq n$. Show $L_n(\varepsilon) = \sigma^{-2} \mathbb{E}[X_1^2 \mathbf{1}\{|X_1| > \varepsilon \sigma \sqrt{n}\}]$ and that it tends to 0. *Dominated convergence, with dominating variable X_1^2 .* Conclude that Theorem 1.9 contains Theorem 1.7 without the third moment.

Exercise 1.14 (TYPE C). Let $X_{n,i}$, $i \leq n$, be independent, equal to ± 1 with probability $\frac{1}{2}(1 - \frac{1}{n})$ each and to $\pm\sqrt{n}$ with probability $\frac{1}{2n}$ each. Show $s_n^2 = 2n - 1$, that Feller's condition holds, and that $L_n(\varepsilon) \rightarrow \frac{1}{2}$ for every $\varepsilon \leq 1/\sqrt{2}$. Then compute $\lim_n \varphi_{T_n}(t)$ and identify the limit law. *Half the variance sits in the ± 1 's and half in the rare $\pm\sqrt{n}$'s; the limit records both. Remark 1.10(c) predicts, before any computation, that the limit cannot be $\mathcal{N}(0, 1)$.*

1.7 Records, and why random search plateaus

Definition 1.8 asks for $s_n \rightarrow \infty$, and that hypothesis is doing more work than it looks.

Lemma 1.15. *If the array of Definition 1.8 is uniformly bounded — $|X_{n,i}| \leq C$ for all n and i — then Lindeberg's condition holds.*

Proof. Fix $\varepsilon > 0$. Since $s_n \rightarrow \infty$ we have $\varepsilon s_n > C$ for all large n , and then the event $\{|X_{n,i}| > \varepsilon s_n\}$ is empty for every i , so $L_n(\varepsilon) = 0$. ■

Here is an array that no central limit theorem for identically distributed summands can reach. Let $X_1, X_2, \dots$ be i.i.d. with a continuous law, so that $\mathbb{P}(X_i = X_j) = 0$ for $i \neq j$, and let

$$B_i = \mathbf{1}\{X_i > \max(X_1, \dots, X_{i-1})\}, \quad B_1 = 1,$$

so that $R_n = \sum_{i \leq n} B_i$ counts the *records* in the first n observations: the times at which the running maximum improves.

Lemma 1.16 (Rényi). *$B_1, \dots, B_n$ are independent, and $\mathbb{P}(B_i = 1) = 1/i$.*

Proof. Let $\rho_i = \#\{j \leq i : X_j \leq X_i\}$ be the rank of X_i among the first i observations, so that $B_i = \mathbf{1}\{\rho_i = i\}$. The vector $(\rho_1, \dots, \rho_n)$ is a function of the relative order of $X_1, \dots, X_n$, which is a uniformly random permutation of $[n]$ because the X_i are i.i.d. and ties do not occur.

The map sending a permutation to $(\rho_1, \dots, \rho_n)$ is a bijection onto $\{1\} \times \{1, 2\} \times \dots \times \{1, \dots, n\}$: given the ρ_i , insert the observations one at a time, placing X_i in position ρ_i among the first i , and this reconstructs the order uniquely. Both sets have $n!$ elements, and a bijection carries the uniform law to the uniform law — which on a product set is the product of uniforms. So the ρ_i are independent with ρ_i uniform on $\{1, \dots, i\}$, and $\mathbb{P}(\rho_i = i) = 1/i$. ■

Theorem 1.17 (Records). *Let $H_n = \sum_{i \leq n} \frac{1}{i}$ and $s_n^2 = H_n - \sum_{i \leq n} \frac{1}{i^2}$. Then*

$$\frac{R_n - H_n}{s_n} \implies \mathcal{N}(0, 1).$$

Proof. Apply Theorem 1.9 to the array $X_{n,i} = B_i - \frac{1}{i}$, $i \leq n$, which is independent by Lemma 1.16 and centred, with $\text{Var}(X_{n,i}) = \frac{1}{i}(1 - \frac{1}{i})$ and hence the stated s_n^2 . Since $s_n^2 \geq H_n - \frac{\pi^2}{6} \rightarrow \infty$ and $|X_{n,i}| \leq 1$, Lemma 1.15 supplies Lindeberg's condition. ■

Nothing here is a normalized sum of i.i.d. summands in disguise. The variances $\frac{1}{i}(1 - \frac{1}{i})$ are not comparable to one another, s_n grows like $\sqrt{\log n}$ rather than $\sqrt{n}$, and no rescaling turns R_n into S_m for any m .

Remark 1.18 ($s_n \rightarrow \infty$ is what excludes the Poisson). Both regimes of §1.6 are arrays of independent Bernoullis; only the variances differ. For $B_{n,i} \sim \text{Bernoulli}(\lambda/n)$, $i \leq n$, the total variance $\lambda(1 - \frac{\lambda}{n})$ stays bounded, Definition 1.8 does not apply at all, and the limit is $\text{Poisson}(\lambda)$. For $B_i \sim \text{Bernoulli}(1/i)$ the total variance diverges, and the limit is Gaussian. The hypothesis $s_n \rightarrow \infty$ is exactly the dial between them.

Remark 1.19 (Random search). Score n hyperparameter configurations drawn independently, and keep the best so far. The number of times the running best improves is precisely R_n , so it is $H_n \approx \log n$ in expectation with fluctuations of order $\sqrt{\log n}$: $\mathbb{E}R_{100} = 5.19$ and $\mathbb{E}R_{10^4} = 9.79$. A

hundredfold increase in budget buys 4.6 further improvements. This is the quantitative content of the observation that random search has diminishing returns, and it is a theorem rather than a heuristic.

Remark 1.20 (A slow theorem). The effective sample size is $s_n^2 \approx \log n$, not n , so the error should be of order $1/s_n$. Exact Kolmogorov–Smirnov distance from $(R_n - H_n)/s_n$ to $\mathcal{N}(0, 1)$:

n	10^2	10^3	10^4	10^5
KS	0.131	0.106	0.089	0.080

against $1/s_n = 0.53, 0.41, 0.35, 0.31$. Halving the error requires raising n to the fourth power. Compare Remark 1.12: R_n is lattice-valued as well, which is the other reason it is slow.

Exercise 1.21 (TYPE A). Let $B_{n,i} \sim \text{Bernoulli}(p_{n,i})$, $i \leq k_n$, be independent and put $\sigma_n^2 = \sum_i p_{n,i}(1 - p_{n,i})$. Show that if $\sigma_n \rightarrow \infty$ then

$$\frac{1}{\sigma_n} \sum_i (B_{n,i} - p_{n,i}) \implies \mathcal{N}(0, 1).$$

One line, given Lemma 1.15. Where does the array of Remark 1.18 with $p_{n,i} = \lambda/n$ fail the hypothesis, and what is the limit there instead?

Exercise 1.22 (TYPE B). Deduce from Theorem 1.17 that $(R_n - \log n)/\sqrt{\log n} \Rightarrow \mathcal{N}(0, 1)$ as well.

First prove the tool: if $Z_n \Rightarrow Z$ with $\sup_n \mathbb{E} Z_n^2 < \infty$, and $a_n \rightarrow 1$, $b_n \rightarrow 0$ are deterministic, then $a_n Z_n + b_n \Rightarrow Z$. For bounded uniformly continuous f , split at $|Z_n| \leq M$ and bound the far piece by $2\|f\|_\infty \mathbb{P}(|Z_n| > M) \leq 2\|f\|_\infty M^{-2} \sup_n \mathbb{E} Z_n^2$. Then use $H_n - \log n \rightarrow \gamma$ and $s_n^2/\log n \rightarrow 1$. Why is Lemma 1.2 not enough on its own here?

Exercise 1.23 (TYPE B). Build a uniformly random permutation of $[n]$ in cycle form by inserting $1, 2, \dots, n$ in turn: element i opens a new cycle with probability $1/i$, and otherwise is placed immediately after one of the $i - 1$ elements already present, each with probability $1/i$. Show that every permutation arises from exactly one sequence of choices and that each sequence has probability $1/n!$, so the construction is uniform. Conclude that the number of cycles K_n has the law of $\sum_{i \leq n} \text{Bernoulli}(1/i)$ with the summands independent, and hence that $(K_n - \log n)/\sqrt{\log n} \Rightarrow \mathcal{N}(0, 1)$.

This is Gončarov’s theorem, and the bijection is the same counting argument as Lemma 1.16.

1.8 * A wide network at initialization

Not lectured. Fix a nonlinearity $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ and let

$$f_m(x) = \frac{1}{\sqrt{m}} \sum_{j=1}^m a_j \sigma(\langle w_j, x \rangle), \quad x \in \mathbb{R}^d,$$

with $w_1, \dots, w_m$ i.i.d. in $\mathbb{R}^d$ and $a_1, \dots, a_m$ i.i.d. with mean 0 and variance σ_a^2 , independent of the w_j . This is a one-hidden-layer network of width m at initialization, normalized so that $\text{Var } f_m(x)$ does not move as the width grows.

Proposition 1.24. *Fix finitely many inputs $x_1, \dots, x_k$ and suppose $\mathbb{E}[\sigma(\langle w, x_i \rangle)^2] < \infty$ for each i . Then*

$$(f_m(x_1), \dots, f_m(x_k)) \implies \mathcal{N}(0, K), \quad K(x, x') = \sigma_a^2 \mathbb{E}[\sigma(\langle w, x \rangle) \sigma(\langle w, x' \rangle)].$$

Exercise 1.25 (TYPE B). Prove it. Put $V_j = (a_j \sigma(\langle w_j, x_1 \rangle), \dots, a_j \sigma(\langle w_j, x_k \rangle)) \in \mathbb{R}^k$ and identify $(f_m(x_1), \dots, f_m(x_k))$ as $m^{-1/2} \sum_j V_j$. Check $\mathbb{E}V_j = 0$ and $\text{Cov}(V_j) = K$, saying where the independence of a_j from w_j is used in each. Then either quote the multidimensional central limit theorem, or apply Theorem 1.7 to the scalars $\langle v, V_j \rangle$ for each fixed $v \in \mathbb{R}^k$, read off convergence of the transform on $\mathbb{R}^k$, and finish with Lemma 1.29. Say which of the two routes needs a third moment.

Exercise 1.26 (TYPE A). Show that K is a valid covariance — that $\sum_{i,l} v_i v_l K(x_i, x_l) \geq 0$ for every $v \in \mathbb{R}^k$ — for any σ with $\mathbb{E}[\sigma(\langle w, x_i \rangle)^2] < \infty$. *One line.*

The limit is Gaussian whatever σ and whatever the law of the weights; those enter only through K .

Exercise 1.27 (TYPE C). Let $\sigma = \max(\cdot, 0)$ and $w \sim \mathcal{N}(0, I_d)$. Show that for $x, x' \neq 0$ with angle $\theta = \angle(x, x')$,

$$\mathbb{E}[\sigma(\langle w, x \rangle) \sigma(\langle w, x' \rangle)] = \frac{\|x\| \|x'\|}{2\pi} (\sin \theta + (\pi - \theta) \cos \theta).$$

Check the answer at $\theta = 0$ against $\mathbb{E}[\max(0, g)^2]$, and at $\theta = \pi/2$ against independence. *This is the arccosine kernel.*

Remark 1.28. Proposition 1.24 concerns finitely many inputs. Holding for every finite subset of $\mathbb{R}^d$ at once is what it means to call the limit a *Gaussian process* with kernel K — Lecture 9. Conditioning it on data is Theorem 2.9 of Lectures 1–2 with Σ replaced by the kernel matrix, so an infinitely wide network fitted by exact Bayesian inference is kernel regression with kernel K . At finite width f_m is not Gaussian, and the size of the discrepancy is a question about the *rate* in Proposition 1.24, which Lecture 4 supplies.

1.9 * The proof we did not give

Not lectured. Characteristic functions turn sums into products, and a product of n nearly-equal factors is an exponential. That is the older proof of Theorem 1.7, and it is worth carrying out once — both to see what it costs, and because the transform does two things at the end of this section that the swap does not.

Lemma 1.29 (Continuity theorem). *Let Y_n and Y be random vectors in $\mathbb{R}^k$ with $\varphi_{Y_n}(t) \rightarrow \varphi_Y(t)$ for every $t \in \mathbb{R}^k$. Then $Y_n \Rightarrow Y$.*

Proof. Let f be continuous with f and $\hat{f}$ both in $L^1(\mathbb{R}^k)$; then f is bounded, being the inverse transform of an integrable function. By Fourier inversion and Fubini, legitimate because $\hat{f} \in L^1$,

$$\mathbb{E}f(Y_n) = \frac{1}{(2\pi)^k} \int_{\mathbb{R}^k} \hat{f}(\xi) \mathbb{E}[e^{i\langle \xi, Y_n \rangle}] d\xi = \frac{1}{(2\pi)^k} \int_{\mathbb{R}^k} \hat{f}(\xi) \varphi_{Y_n}(-\xi) d\xi.$$

Since $|\varphi_{Y_n}| \leq 1$ and $\hat{f} \in L^1$, dominated convergence gives $\mathbb{E}f(Y_n) \rightarrow \frac{1}{(2\pi)^k} \int \hat{f}(\xi) \varphi_Y(-\xi) d\xi = \mathbb{E}f(Y)$.

The functions used in Exercise 1.3 all lie in this class: a smooth cutoff χ_M is compactly supported, so $\widehat{\chi_M}$ is Schwartz; and $g_\delta = g * \phi_\delta$ is integrable with $\hat{g}_\delta = \hat{g} \hat{\phi}_\delta$ a bounded function times a Gaussian. That argument therefore applies verbatim and gives $Y_n \Rightarrow Y$. (The imported statements of Lectures 1–2 are all in $\mathbb{R}^k$.) ■

Exercise 1.30 (TYPE A). For $\mathbb{E}|X|^k < \infty$, differentiate under the expectation k times — exhibit the dominating variable — to get $\varphi_X^{(k)}(0) = (-i)^k \mathbb{E}[X^k]$. Check it against $\varphi(t) = e^{-t^2/2}$ by computing $\varphi''(0)$ and $\varphi^{(4)}(0)$, recovering $\mathbb{E}g^2 = 1$ and $\mathbb{E}g^4 = 3$.

Lemma 1.31. If $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$ then $\varphi_X(t) = 1 - \frac{t^2}{2} + r(t)$ with $r(t)/t^2 \rightarrow 0$.

Proof. For real θ , $|e^{i\theta} - 1 - i\theta + \frac{\theta^2}{2}| \leq \min\left(\frac{|\theta|^3}{6}, \theta^2\right)$: the first bound is the integral form of the Taylor remainder, the second follows from $|e^{i\theta} - 1 - i\theta| \leq \frac{\theta^2}{2}$. Put $\theta = -tX$ and take expectations, using $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$:

$$|r(t)| \leq \mathbb{E} \min\left(\frac{|t|^3|X|^3}{6}, t^2 X^2\right).$$

Divide by t^2 . The integrand is bounded by X^2 , which is integrable, and tends to 0 pointwise, so dominated convergence gives $r(t)/t^2 \rightarrow 0$. The second branch of the minimum is what supplies the dominating variable without a third moment. ■

Exercise 1.32 (TYPE B). Show $(1+z_n/n)^n \rightarrow e^z$ whenever $z_n \rightarrow z$ in $\mathbb{C}$ (for $|a|, |b| \leq M$, $|a^n - b^n| \leq nM^{n-1}|a - b|$; take $b = e^{z_n/n}$). Combine with Lemma 1.31 to get $\varphi_{S_n}(t) = [\varphi_{X_1}(t/\sqrt{n})]^n \rightarrow e^{-t^2/2}$ and conclude $S_n \Rightarrow \mathcal{N}(0, 1)$. Note where independence was used, and where identical distribution was used.

Exercise 1.33 (TYPE A). Let $B \sim \text{Bernoulli}(p)$; show $\varphi_B(t) = 1 + p(e^{-it} - 1)$. For independent $B_{n,i} \sim \text{Bernoulli}(\lambda/n)$ with $N_n = \sum_i B_{n,i}$, show $\varphi_{N_n}(t) \rightarrow \exp(\lambda(e^{-it} - 1))$ and identify the limit as $\text{Poisson}(\lambda)$ by summing the series. *The same three lines as Exercise 1.32; only the normalization differs.*

Exercise 1.34 (TYPE B). For the centred array $X_{n,i} = B_{n,i} - \lambda/n$, show $s_n^2 = \lambda(1 - \lambda/n)$ and that $L_n(\varepsilon) \rightarrow 1$ for every $\varepsilon < 1/\sqrt{\lambda}$, so Definition 1.8 fails. Compute $L_n(\frac{1}{2})$ exactly at $\lambda = 3$ for $n = 10, 100, 10^4$; you should get 0, 0.970, 1.000, and the first should be exactly 0 — say why.

Exercise 1.35 (TYPE C). Take $X_{n,i} = \pm 1$ for $i \leq n$ and one extra summand $X_{n,n+1} = \pm\sqrt{n}$, all independent and symmetric. Show $s_n^2 = 2n$, that $L_n(\varepsilon) \geq \frac{1}{2}$ for $\varepsilon < 1/\sqrt{2}$, and that the limiting characteristic function of $s_n^{-1} \sum_i X_{n,i}$ is $e^{-t^2/4} \cos(t/\sqrt{2})$. Exhibit a t at which it vanishes, and conclude that $s_n^{-1} \sum_i X_{n,i}$ converges in distribution to no Gaussian at all. *Feller's condition fails here as well, so Remark 1.10(c) does not apply and the transform must be computed. Why does one zero settle it, without knowing in advance that a limit exists?*

Exercise 1.36 (TYPE C). The Berry–Esseen bound of Remark 2.9 gives rate $n^{-1/2}$ in Kolmogorov distance. Show that for $X_i = \pm 1$ it cannot be improved. For n even, $\mathbb{P}(S_n = 0) = \binom{n}{n/2} 2^{-n}$, which Stirling puts at $\sim \sqrt{2/\pi n}$; now compare $\mathbb{P}(S_n \leq z)$ with $\Phi(z)$ near $z = 0$. Explain in one sentence why lattice summands were the slowest row of the table in §1.6.

Exercise 1.37 (TYPE C). Show that for every real random variable Y and every $u > 0$,

$$\mathbb{P}(|Y| \geq 2/u) \leq \frac{1}{u} \int_{-u}^u (1 - \varphi_Y(t)) dt.$$

Deduce that if $\varphi_{Y_n} \rightarrow \psi$ pointwise with ψ continuous at 0, then $\sup_n \mathbb{P}(|Y_n| > K) \rightarrow 0$ as $K \rightarrow \infty$. *This is the half of the continuity theorem that does not need the limit law in advance. The other half — that ψ is then the transform of some random variable — is a compactness argument we do not need, since Lemma 1.29 is always applied with the limit already in hand.*

ASSIGNED Lecture 3, due at the start of Lecture 4: Exercises 1.3, 1.13, 1.14, 1.21, 1.22, 1.25, 1.32, 1.35 — two A, three B, three C. Every other exercise in §1 is a pool problem: available and examinable, not collected. *If you are short of time, do the C problems.*

2 Stein's method

Theorem 1.7 describes the Gaussian as a limit. There is another way to pin it down.

Q. Is there an identity satisfied by the standard Gaussian and by no other law? If so, then showing that a law is approximately Gaussian becomes showing that it approximately satisfies that identity.

2.1 The characterization

Theorem 2.1 (Stein's lemma). *A random variable W is standard Gaussian if and only if*

$$\mathbb{E}[Wf(W)] = \mathbb{E}[f'(W)]$$

for every bounded absolutely continuous f with bounded derivative.

Proof of the forward direction. Let $\phi(x) = (2\pi)^{-1/2}e^{-x^2/2}$, so that $\phi'(x) = -x\phi(x)$. Then

$$\mathbb{E}[f'(g)] = \int f' \phi = - \int f \phi' = \int x f(x) \phi(x) dx = \mathbb{E}[gf(g)].$$

■

The converse is proved in §2.2.

Proposition 2.2 (Vector form). *Let $X \sim \mathcal{N}(0, \Sigma)$ on $\mathbb{R}^d$ and let $f \in C^1(\mathbb{R}^d)$ with f and ∇f of at most polynomial growth. Then*

$$\mathbb{E}[X_i f(X)] = \sum_j \Sigma_{ij} \mathbb{E}[\partial_j f(X)], \quad \text{that is} \quad \mathbb{E}[Xf(X)] = \Sigma \mathbb{E}[\nabla f(X)].$$

Conversely, if a random vector W on $\mathbb{R}^d$ satisfies

$$\mathbb{E}[\langle W, F(W) \rangle] = \mathbb{E}[\text{tr}(\Sigma DF(W))]$$

for every bounded $F : \mathbb{R}^d \rightarrow \mathbb{R}^d$ with bounded derivative, then $W \sim \mathcal{N}(0, \Sigma)$.

Proof. Write $X = AZ$ with $AA^\top = \Sigma$ and $Z \sim \mathcal{N}(0, I_d)$, which is legitimate by Theorem 1.5 and Proposition 1.12 of Lectures 1–2. The coordinates of Z are independent standard Gaussians, so conditioning on the others and applying the identity of Theorem 2.1 in the k th gives $\mathbb{E}[Z_k h(Z)] = \mathbb{E}[\partial_k h(Z)]$ for $h \in C^1$ of polynomial growth — the integration by parts is the one already done, and its boundary terms vanish because ϕ decays faster than any polynomial. Apply this to $h = f \circ A$:

$$\mathbb{E}[X_i f(X)] = \sum_k A_{ik} \mathbb{E}[Z_k (f \circ A)(Z)] = \sum_k A_{ik} \mathbb{E}[\partial_k (f \circ A)(Z)] = \sum_j \left(\sum_k A_{ik} A_{jk} \right) \mathbb{E}[\partial_j f(X)],$$

and the inner sum is Σ_{ij} .

For the converse, fix $v \in \mathbb{R}^d$ and take $F(x) = v f(\langle v, x \rangle)$ with f bounded and f' bounded. Then $\langle W, F(W) \rangle = Y f(Y)$ for $Y = \langle v, W \rangle$, while $DF(x) = f'(\langle v, x \rangle) v v^\top$, so $\text{tr}(\Sigma DF) = \sigma^2 f'(Y)$ with $\sigma^2 = v^\top \Sigma v$. The hypothesis therefore reads $\mathbb{E}[Y f(Y)] = \sigma^2 \mathbb{E}[f'(Y)]$ for every such f . If $\sigma > 0$, this is the identity of Theorem 2.1 for Y/σ , so $Y \sim \mathcal{N}(0, \sigma^2)$; if $\sigma = 0$, then $f(y) = y/(1 + y^2)$ is admissible and gives $\mathbb{E}[Y^2/(1 + Y^2)] = 0$, so $Y = 0$. Every linear functional of W is thus $\mathcal{N}(0, v^\top \Sigma v)$, and $W \sim \mathcal{N}(0, \Sigma)$ by Corollary 1.7 of Lectures 1–2. ■

Proposition 2.3 (Matrix form). *Let M be a random matrix whose entries are jointly Gaussian with mean zero, and let F be C^1 in the entries with F and its partial derivatives of at most polynomial growth. Then*

$$\mathbb{E}[M_{ij} F(M)] = \sum_{k,l} \mathbb{E}[M_{ij} M_{kl}] \mathbb{E}\left[\frac{\partial F}{\partial M_{kl}}(M)\right].$$

Proof. Proposition 2.2 with the index set $\{(k, l)\}$ in place of $\{1, \dots, d\}$: a matrix is a vector whose index happens to be a pair, and its covariance matrix is $\Sigma_{(ij), (kl)} = \mathbb{E}[M_{ij} M_{kl}]$. ■

Remark 2.4 (Where these are used). Taking $f(x) = x_{a_2} \cdots x_{a_m}$ in Proposition 2.2, so that $\partial_j f$ deletes one factor, gives

$$\mathbb{E}[X_{a_1} X_{a_2} \cdots X_{a_m}] = \sum_{s=2}^m \Sigma_{a_1 a_s} \mathbb{E}\left[\prod_{r \neq 1, s} X_{a_r}\right],$$

the recursion that proves Wick's theorem in Lectures 5–6.

The matrix form is the standard route to the limiting spectrum of a Gaussian random matrix. With $G = (M - z)^{-1}$ one has $\partial G_{ab}/\partial M_{kl} = -G_{ak} G_{lb}$, so Proposition 2.3 turns $\mathbb{E}[M_{ij} G_{ab}]$ into a sum of products of resolvent entries and yields a closed equation for $\frac{1}{N} \mathbb{E} \text{tr} G$. Lectures 5–6 derive that equation by a Schur complement instead, and the reason is visible here: this route uses the joint Gaussianity of the entries, and the answer does not depend on it.

Remark 2.5 (Already used). With $f(x) = e^{-itx}$, Stein's identity becomes

$$\mathbb{E}[W e^{-itW}] = -it \mathbb{E}[e^{-itW}],$$

that is $\varphi'(t) = -t\varphi(t)$, the differential equation of Lemma 1.10 of Lectures 1–2. The first computation of the course was Stein's lemma with an exponential test function.

Splitting off one summand and Taylor-expanding to reach $\varphi'_{S_n} \approx -t\varphi_{S_n}$ proves the central limit theorem; that route is §1.9*. Exponential test functions give no rate in a probability metric and no tolerance for dependence.

2.2 Stein's equation

Fix a test function h and look for f solving

$$f'(w) - wf(w) = h(w) - \mathbb{E}h(g). \quad (3)$$

Throughout this section $h \in C^3(\mathbb{R})$ with h', h'' and h''' bounded — the class of Lemma 1.2, so that control of $\mathbb{E}h(W) - \mathbb{E}h(g)$ over it is control of the law of W .

Lemma 2.6 (The solution, and its derivatives). *Equation (3) has exactly one bounded solution, namely*

$$f_h(w) = - \int_0^\infty e^{-s} \mathbb{E} \left[h'(e^{-s}w + \sqrt{1 - e^{-2s}} g) \right] ds,$$

and it satisfies

$$\|f_h\|_\infty \leq \|h'\|_\infty, \quad \|f'_h\|_\infty \leq \tfrac{1}{2} \|h''\|_\infty, \quad \|f''_h\|_\infty \leq \tfrac{1}{3} \|h'''\|_\infty.$$

Proof. Put $b_s = \sqrt{1 - e^{-2s}}$ and

$$u(s, w) = \mathbb{E} [h(e^{-s}w + b_s g)],$$

so that $u(0, w) = h(w)$ and $u(s, w) \rightarrow \mathbb{E}h(g)$ as $s \rightarrow \infty$. Differentiating in s , and using $\dot{b}_s = e^{-2s}/b_s$,

$$\partial_s u = \mathbb{E} \left[h'(e^{-s}w + b_s g) \left(-e^{-s}w + \frac{e^{-2s}}{b_s} g \right) \right] = -e^{-s}w \mathbb{E} [h'] + \frac{e^{-2s}}{b_s} \mathbb{E} [g h'].$$

Theorem 2.1, applied to $g \mapsto h'(a + b_s g)$, gives $\mathbb{E} [g h'(a + b_s g)] = b_s \mathbb{E} [h''(a + b_s g)]$, which also removes the apparent singularity at $s = 0$. Since $\partial_w u = e^{-s} \mathbb{E} [h']$ and $\partial_w^2 u = e^{-2s} \mathbb{E} [h'']$, this says exactly

$$\partial_s u = \partial_w^2 u - w \partial_w u.$$

Now $f_h(w) = - \int_0^\infty \partial_w u(s, w) ds$ is the displayed formula, and

$$f'_h(w) - w f_h(w) = - \int_0^\infty (\partial_w^2 u - w \partial_w u) ds = - \int_0^\infty \partial_s u ds = u(0, w) - \lim_{s \rightarrow \infty} u(s, w) = h(w) - \mathbb{E}h(g),$$

so f_h solves (3). Two solutions differ by a solution of $f' = wf$, that is by $ce^{w^2/2}$, which is bounded only for $c = 0$; and f_h is bounded, by $\|h'\|_\infty \int_0^\infty e^{-s} ds$. Differentiating under the integral — legitimate since the k th integrand is dominated by $e^{-ks} \|h^{(k)}\|_\infty$ — gives

$$f'_h(w) = - \int_0^\infty e^{-2s} \mathbb{E} [h''(e^{-s}w + b_s g)] ds, \quad f''_h(w) = - \int_0^\infty e^{-3s} \mathbb{E} [h'''(e^{-s}w + b_s g)] ds,$$

and $\int_0^\infty e^{-ks} ds = 1/k$. ■

Remark 2.7. Solving (3) as a linear first-order equation instead gives the classical closed form $f_h(w) = e^{w^2/2} \int_{-\infty}^w [h(x) - \mathbb{E}h(g)] e^{-x^2/2} dx$. It is the same function, by uniqueness. The representation above is the one that differentiates, because there the w -dependence sits inside h and every derivative costs a factor e^{-s} .

Evaluating (3) at W and taking expectations turns a distance between two laws into a single expectation:

$$\mathbb{E}h(W) - \mathbb{E}h(g) = \mathbb{E} [f'_h(W) - W f_h(W)]. \quad (4)$$

This settles the converse in Theorem 2.1. If W satisfies the identity, then for every h in our class f_h is bounded with bounded derivative by Lemma 2.6, hence an admissible test function, so the right side of (4) vanishes; thus $\mathbb{E}h(W) = \mathbb{E}h(g)$ for every such h , and $W \stackrel{d}{=} g$ by Lemma 1.2 and Exercise 1.3.

2.3 The central limit theorem, with a rate

Theorem 2.8. *Let $X_1, \dots, X_n$ be i.i.d. with mean 0, variance 1 and $\mathbb{E}|X_1|^3 < \infty$. Then for every $h \in C^3(\mathbb{R})$ with bounded derivatives,*

$$|\mathbb{E}h(S_n) - \mathbb{E}h(g)| \leq \frac{\|h'''\|_\infty \mathbb{E}|X_1|^3}{2\sqrt{n}}.$$

Proof. Write $W = S_n$, let $W^{(i)} = W - X_i/\sqrt{n}$, which is independent of X_i , and abbreviate $f = f_h$. Then

$$\mathbb{E}[Wf(W)] = \frac{1}{\sqrt{n}} \sum_i \mathbb{E}[X_i f(W)],$$

and Taylor expanding $f(W) = f(W^{(i)}) + \frac{X_i}{\sqrt{n}} f'(W^{(i)}) + R_i$ with $|R_i| \leq \frac{\|f''\|_\infty}{2} \frac{X_i^2}{n}$ gives, for each i ,

$$\mathbb{E}[X_i f(W)] = \underbrace{\mathbb{E}[X_i] \mathbb{E}[f(W^{(i)})]}_{=0} + \frac{1}{\sqrt{n}} \underbrace{\mathbb{E}[X_i^2]}_{=1} \mathbb{E}[f'(W^{(i)})] + \mathbb{E}[X_i R_i],$$

where $|\mathbb{E}[X_i R_i]| \leq \|f''\|_\infty \mathbb{E}|X_1|^3/2n$ and the first two expectations factor because X_i is independent of $W^{(i)}$. Summing,

$$\left| \mathbb{E}[Wf(W)] - \frac{1}{n} \sum_i \mathbb{E}[f'(W^{(i)})] \right| \leq \frac{\|f''\|_\infty \mathbb{E}|X_1|^3}{2\sqrt{n}}.$$

Each $\mathbb{E}[f'(W^{(i)})]$ differs from $\mathbb{E}[f'(W)]$ by at most $\|f''\|_\infty \mathbb{E}|X_i|/\sqrt{n}$, so

$$|\mathbb{E}[f'(W) - Wf(W)]| \leq \frac{\|f''\|_\infty}{\sqrt{n}} \left(\mathbb{E}|X_1| + \frac{1}{2} \mathbb{E}|X_1|^3 \right) \leq \frac{3\|f''\|_\infty \mathbb{E}|X_1|^3}{2\sqrt{n}},$$

since $\mathbb{E}|X_1| \leq (\mathbb{E}X_1^2)^{1/2} = 1 \leq (\mathbb{E}X_1^2)^{3/2} \leq \mathbb{E}|X_1|^3$. Now insert $\|f''\|_\infty \leq \frac{1}{3}\|h'''\|_\infty$ from Lemma 2.6 into (4). $\blacksquare$

Remark 2.9 (Nothing has been won yet, and what wins it). Theorem 1.7 gave $\frac{\|h'''\|_\infty}{6\sqrt{n}}(\mathbb{E}|X_1|^3 + 2\sqrt{2/\pi})$ on the same class: the swap and Stein's method agree, to a constant. The gain comes from measuring h in $\|h'\|_\infty$ rather than $\|h'''\|_\infty$. The bound below, valid for every Lipschitz h , turns the display in the proof above into

$$|\mathbb{E}h(S_n) - \mathbb{E}h(g)| \leq \frac{3\|h'\|_\infty \mathbb{E}|X_1|^3}{\sqrt{n}},$$

a bound on the Wasserstein distance between the two laws; approximating $\mathbf{1}_{(-\infty, z]}$ by Lipschitz functions and controlling the approximation then upgrades it to the Berry–Esseen theorem, $\sup_z |\mathbb{P}(S_n \leq z) - \mathbb{P}(g \leq z)| \leq C \mathbb{E}|X_1|^3/\sqrt{n}$. The extra work is in the indicator, not in the method.

Stated without proof. If h is Lipschitz then f_h is twice differentiable with $\|f_h''\|_\infty \leq 2\|h'\|_\infty$. This is the only ingredient of §2.2–§2.3 we do not prove; the representation of Lemma 2.6 gives $\|f_h''\|_\infty \leq \frac{1}{3}\|h'''\|_\infty$ in one line but not the Lipschitz version, which needs the closed form and a careful splitting.

What Theorem 2.8 does buy outright is everything the swap could not reach: the same machinery with a different characterizing identity gives a Poisson limit theorem with a rate in total variation (§2.4), and it survives the loss of independence (§2.5*).

2.4 The same argument for Poisson

Nothing above was specific to the Gaussian; only the characterizing identity was used.

Theorem 2.10 (Chen's lemma). *A random variable N taking values in $\{0, 1, 2, \dots\}$ is $\text{Poisson}(\lambda)$ if and only if*

$$\mathbb{E}[\lambda f(N+1) - Nf(N)] = 0$$

for every bounded f .

Proof of the forward direction. $\mathbb{E}[Nf(N)] = \sum_{k \geq 1} k e^{-\lambda} \frac{\lambda^k}{k!} f(k) = \lambda \sum_{j \geq 0} e^{-\lambda} \frac{\lambda^j}{j!} f(j+1) = \lambda \mathbb{E}[f(N+1)]$. ■

Running §2.2 and §2.3 with this identity in place of Stein's, and total variation in place of a smooth metric, gives:

Theorem 2.11 (Le Cam). *Let $B_1, \dots, B_n$ be independent with $B_i \sim \text{Bernoulli}(p_i)$, and let $\lambda = \sum_i p_i$. Then*

$$d_{\text{TV}}\left(\sum_i B_i, \text{Poisson}(\lambda)\right) \leq \sum_i p_i^2.$$

Corollary 2.12 (Poisson limit theorem). *Let $B_{n,i} \sim \text{Bernoulli}(p_{n,i})$, $i \leq k_n$, be independent, with $\max_i p_{n,i} \rightarrow 0$ and $\sum_i p_{n,i} \rightarrow \lambda$. Then $\sum_i B_{n,i} \Rightarrow \text{Poisson}(\lambda)$.*

Proof. $\sum_i p_{n,i}^2 \leq (\max_i p_{n,i}) \sum_i p_{n,i} \rightarrow 0$, and $\text{Poisson}(\lambda_n) \Rightarrow \text{Poisson}(\lambda)$ when $\lambda_n \rightarrow \lambda$. ■

This is the Poisson side of the Lindeberg boundary. At $p_i = \lambda/n$ the rate is λ^2/n ; with $\lambda = 3$ and $n = 100$ the bound is 0.09, against an actual discrepancy of about 0.002 in $\mathbb{P}(\sum_i B_i = 0)$.

Lecture 3 met the Gaussian/Poisson boundary as a phenomenon. Here a single technique sits on both sides of it, and only the characterizing identity changes.

2.5 * Where independence was spent

Not lectured. The proof of Theorem 2.8 used independence in exactly one place, to evaluate $\mathbb{E}[X_i f(W^{(i)})] = \mathbb{E}[X_i] \mathbb{E}[f(W^{(i)})]$. Any other route to that expectation will do, and the cheapest is a coupling.

Definition 2.13 (Exchangeable pair). A pair (W, W') is exchangeable if $(W, W') \stackrel{d}{=} (W', W)$. It is a *Stein pair* if in addition $\mathbb{E}[W' - W \mid W] = -aW$ for some $a > 0$.

Exercise 2.14 (TYPE A). Let (W, W') be exchangeable and F antisymmetric, $F(x, y) = -F(y, x)$, with $\mathbb{E}|F(W, W')| < \infty$. Show $\mathbb{E}F(W, W') = 0$. *One line, and it is the whole mechanism; everything else is choosing F .*

Exercise 2.15 (TYPE B). Apply that to $F(x, y) = (y - x)(f(x) + f(y))$, expand $f(W')$ about W to first order, and use Definition 2.13 to obtain

$$\mathbb{E}[f'(W) - Wf(W)] = \mathbb{E}\left[f'(W)\left(1 - \frac{1}{2a}\mathbb{E}[(W' - W)^2 \mid W]\right)\right] + \text{err}, \quad |\text{err}| \leq \frac{\|f''\|_\infty}{4a} \mathbb{E}|W' - W|^3.$$

A Gaussian approximation now asks two things: that $\frac{1}{2a}\mathbb{E}[(W' - W)^2 \mid W]$ be close to 1, and that $W' - W$ be small.

Exercise 2.16 (TYPE B). *The canonical pair.* With $W = S_n$, let I be uniform on $\{1, \dots, n\}$ and X'_I an independent copy of X_I ; put $W' = W - (X_I - X'_I)/\sqrt{n}$. Show (W, W') is exchangeable with $\mathbb{E}[W' - W \mid X_1, \dots, X_n] = -W/n$, so $a = 1/n$; then that $\mathbb{E}[(W' - W)^2 \mid X_1, \dots, X_n] = n^{-2} \sum_i (X_i^2 + 1)$, whose distance from 1 after dividing by $2a$ is at most $\frac{1}{2} \sqrt{\text{Var}(X_1^2)}/n$. Conclude Theorem 2.8 again. *Independence was used twice; neither use was the product formula.*

Now a sum to which no version of independence applies. Let π be a uniformly random permutation of $\{1, \dots, n\}$, let $I_i = \mathbf{1}\{\pi(i) = i\}$, and let $W = \sum_i I_i$ count fixed points.

Exercise 2.17 (TYPE A). Show $\mathbb{E}W = \text{Var}W = 1$ for $n \geq 2$, and that $\mathbb{P}(I_1 = I_2 = 1) \neq \mathbb{P}(I_1 = 1)\mathbb{P}(I_2 = 1)$. A student applies Theorem 2.11 with $p_i = 1/n$ and $\lambda = 1$ to conclude $d_{\text{TV}}(W, \text{Poisson}(1)) \leq 1/n$. Name the hypothesis that fails, and say why the conclusion happening to be true does not rescue the argument.

Imported: Chen–Stein, coupling form. Let $W = \sum_{i \leq n} I_i$ with $I_i \sim \text{Bernoulli}(p_i)$, *not necessarily independent*, and $\lambda = \sum_i p_i$. If for each i one can build, on the same space as W , a variable V_i distributed as $W - 1$ conditioned on $I_i = 1$, then $d_{\text{TV}}(W, \text{Poisson}(\lambda)) \leq \min(1, \lambda^{-1}) \sum_i p_i \mathbb{E}|W - V_i|$.

Exercise 2.18 (TYPE C). Build the coupling. Fix i ; if $\pi(i) = i$ put $\sigma = \pi$, otherwise $\sigma = \pi \circ (i \ \pi^{-1}(i))$. Show $\sigma(i) = i$ always, that σ is uniform among permutations fixing i , and that $W(\sigma) - W(\pi) \in \{1, 2\}$ when $\pi(i) \neq i$ — the value 2 exactly when π has a 2-cycle at i . Deduce $\mathbb{E}|W - V_i| \leq 2/n$, hence $d_{\text{TV}}(W, \text{Poisson}(1)) \leq 2/n$.

Inclusion–exclusion gives $\mathbb{P}(W = 0) = \sum_{k \leq n} (-1)^k / k!$, whose distance from e^{-1} is 7.1×10^{-3} , 1.8×10^{-4} and 2.3×10^{-8} at $n = 4, 6, 10$. *By how many orders of magnitude is the method wrong at $n = 10$, and what does it have that inclusion–exclusion does not?*

§1.8* carries out the wide-network limit and computes the arccosine kernel; §1.9* gives the characteristic-function proof of the central limit theorem, the same computation for Poisson, and Lindeberg’s condition on both sides of the boundary; §2.5* develops the exchangeable-pair bound and a case with no independence at all.

ASSIGNED Lecture 4, due at the start of Lecture 5: every exercise in §2. *If you are short of time, do the C problem.*